

Aaron Lam

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EXPERIENCE

Embedded System Software Engineer | TP-Link Systems Inc. | Irvine, CA

June 2025 – Present

Working on smart home cameras and doorbells. Use **C**, **RTOS**, and **Embedded Linux** to implement new features and fix bugs within proprietary libraries. Debugged communication issues across various platforms including MQTT and HTTP cloud servers as well as smart home hubs. Lead firmware release for QA test cycle including creating and shipping firmware release builds along with filling out required documentation.

Software Engineer Level 2 | Veranex/Fusion Biotech Inc.

April 2023 – June 2025

Worked on medical equipment, and performed the following software activities:

- Developed embedded touch display driver and associated user interface using **C/C++**, **STM32**, and **STM32 TouchGFX**
 - Created user interface to add various UI components like interactive buttons, spin boxes, etc. for different screens for the touch display application
 - Developed driver for user interaction using touch controls
 - Created interface to save/upgrade assets (images, fonts, etc.) in non-volatile memory and have MCU read/use saved assets
 - Designed and integrated communication so that MCU used for touch display and application can communicate to separate MCU which handles actions like controlling motors and heaters
- Performed verification testing to validate a medical device using **C** and **Python**
 - Created test scripts using propriety communication protocol to run various test steps and generate a test report
 - Setup and ran code safety and security checks like static code analysis, unit tests, etc.
 - Wrote and presented/shared reports regarding warnings and issues
 - Integrated test scripts with hardware in the loop (HIL) system.
 - Communicated with client in Switzerland and guided engineers in using scripts
- Developed prototype Bluetooth implantable device and associated iOS mobile app using **C/C++** and **Swift**
 - Developed implantable to be able to communicate via Bluetooth, control a motor, self-monitor battery life, and handle sleep/wake up to preserve battery life
 - Created state machines for different states of implantable
 - Developed iOS mobile app that controls implantable like starting motor actions and reading device data over Bluetooth
 - Created initial prototype of app and initial draft of user interface for UI designers to reference/improve upon
- Designed various drivers in **C/C++** for different components including NFC reader/writer, heaters, motors, fuel gauge, touch display, buttons (including haptic touch), LEDs, and memory (EEPROM, non-volatile, and flash)
 - Implemented different embedded protocols like UART, SPI, ADC, GPIO, PWM, and DAC to interact with components
 - Resolved issues using Debuggers, oscilloscopes, signal analyzers, and voltmeters
- Developed IOT driver for STM32 board to communicate to MQTT server using **C/C++** and **Docker**
 - Created Docker image that hosts a MQTT server in Linux environment with custom DNS and certificates
 - Integrated MQTT driver so that STM32 board can communicate to the Docker hosted MQTT server
- Developed full stack for oxygen tracking mobile app with 1000+ users using **Dart** and **Flutter**, **TypeScript**, **ReactJS**, and **NextJS**
 - Worked on various graphing modules in the web application. Handled pagination and associated viewing, sorting, filtering, and retrieving users and oxygen data.
 - Integrated communication with Bluetooth oxygen ring so that the app can interface with the oxygen ring to read, collect, and save oxygen data
 - Set up internal storage and caching of data to allow users to access data quicker and as a backup
 - Developed various UI screens: User login, user and device onboarding, device management, and oxygen statistic graphs, drop down selection, buttons, background, animations, etc.
- Debugged and assisted with cloud server used for web and mobile applications with 1000+ users using **Python** and **FastAPI**

Worked on medical equipment, and performed the following software activities:

- Developed internal diagnostics and safe state handling system for medical devices in C/C++
- Implemented features like voltage monitoring, temperature checking, and flash integrity checking, put the device into a safe state to prevent any usage that could cause damage to the device
- Updated communication between embedded devices and applications to allow for multiple devices to be communicated to at the same time using C/C++ and Python with PyQt.
- Implemented encrypted communication and key exchange between applications and embedded devices using C/C++ and Python with PyQt.

Intern Software Engineer | Veranex/Fusion Biotech Inc.

November 2021 – April 2022

Worked on medical equipment, and performed the following software activities under the guidance of a senior engineer:

- Added AES encryption and integrated encryption/decryption of messages between medical device and application
- Developed custom handshaking and key exchange protocol between medical device and application

Dorm Staff | University of California - Irvine

September 2019 – March 2020

Assisted freshman with academic resources and planned social events within the dorm halls

VOLUNTEER

Tutor and Learning Assistant

January 2021 – June 2021

Assisted students in Introduction to Programming, Principles of Systems Design, Boolean Algebra, and Discrete Mathematics classes. Attended lectures to help assist students with activities and questions. Held office hours for students to be able to get additional assistance.

SKILLS

Programming Languages

C, C++, Python, Dart, TypeScript, JavaScript/JSX, HTML, CSS, SQL, Rust, Unix/Bash, Java, C#, Swift

Frameworks & Libraries

Qt/PyQt, Flutter, React, Angular, STM32, JQuery, NodeJS, Flask, SocketIO, PyGame

Operating Systems

Windows, Linux, Mac, iOS, Android

Software & Infrastructures

Git, Docker, JIRA, Figma

EDUCATION

University of California, Irvine

September 2018 – December 2021

Bachelor of Science in Computer Science | Specialization in Visual Computing | Dean's Honor List

GPA: 3.762

REFERENCES

Will be provided upon request